

Determination of ^{210}Pb by liquid scintillation counting of ^{210}Pb and its progenies

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Introduction

- Characteristics of ^{210}Pb

- ^{210}Pb ($\tau_{1/2}$: 22.3 yr) is one of the nuclides of the ^{238}U decay series and decays to ^{210}Bi ($\tau_{1/2}$: 5 d) by emitting β and γ rays, which decays to ^{210}Po ($\tau_{1/2}$: 138 d).
- ^{210}Pb is used as tracers for food chain studies and in age-dating of environmental materials, because it is widely distributed in soil, sediments, atmosphere and water, etc.

- Measurement of ^{210}Pb

- There are several methods to analyze ^{210}Pb
 - indirect measurements of ^{210}Pb by an alpha spectrometry or ^{210}Bi by an α/β proportional counter
 - direct measurement of ^{210}Pb using a gamma spectrometry
- several limits (much time / low efficiency / self-absorption)
- More rapid measurement by Liquid scintillation counter (LSC)

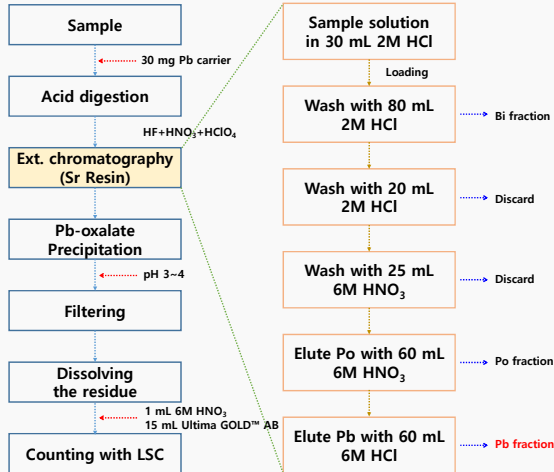
- Study Objectives

- To evaluate 1-Window method which detects the counts of whole energy regions (^{210}Pb + ^{210}Bi + ^{210}Po) as ^{210}Pb for high efficiency in low-level environmental materials
- To verify a calculation of ^{210}Pb activities including blank value from Pb^{2+} carrier

Experimental

- Extraction Chromatography

[Extraction chromatography]
- Sr Resin -



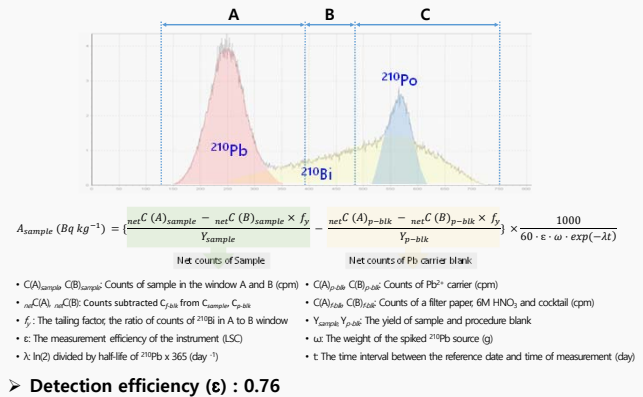
- Measurement

- In order to form a precipitate, 0.3 g oxalic acid is added in the warm Pb fraction and the pH is adjusted to 3-5 with NH_4OH .
- Following the filtration, the precipitate is dried in an oven at 50 °C for 90 min and dissolved with 1 mL 6M HNO_3 and mixed with 15 mL cocktail (Ultima GOLD™ AB) in a small PFA vial (20 mL).
- The samples are measured for 300 min by a liquid scintillation counter (1220 Quantulus EG & G Wallac Oy).

Results & Discussion

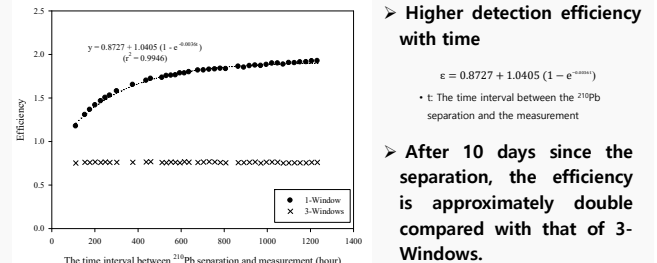
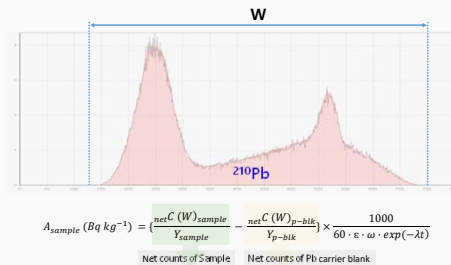
- Determination of ^{210}Pb by 3-Windows method

- The A window is composed with ^{210}Pb beta peak and a low energy region of ^{210}Bi beta peak. The B window is only ^{210}Bi middle region and the C window is the sum of ^{210}Bi in the high energy region and ^{210}Po alpha peak.
- Blank correction was performed by using the value of the procedure blank derived from Pb^{2+} carrier in order to determine the accurate value.
- It is necessary to consider the ^{210}Pb ingrowth of carrier when using carrier gravimetric method to obtain the chemical yield.



- Determination of ^{210}Pb by 1-Window method (Improved method)

- It regards whole windows as the activities of ^{210}Pb , instead of dividing the three windows depending on the energy of the three nuclides.



➤ Higher detection efficiency with time

$$\epsilon = 0.8727 + 1.0405(1 - e^{-0.0003t})$$

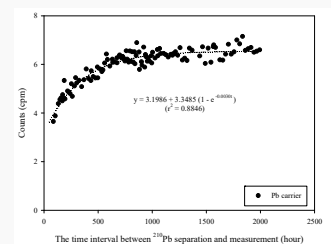
• t: The time interval between the ^{210}Pb separation and the measurement

- After 10 days since the separation, the efficiency is approximately double compared with that of 3- Windows.

- The counts of procedure blank, Pb^{2+} carrier, also increase with time.

$$\frac{\text{netC}(W)_{p-blk}}{Y_{p-blk}} = 3.1986 + 3.3485(1 - e^{-0.0030t})$$

- This method was verified by using the certified reference material (IAEA-447) and was in good agreement within 1% of the certified value.



IAEA-447	Certified Value	1-Window	3-Windows
^{210}Pb value (Bq kg^{-1})	420 ± 20	416 ± 9.8* (n = 32)	427 ± 15* (n = 32)
Minimum detectable activity (Bq kg^{-1})	-	18**	33**

* Standard deviation ** After 10 days since the ^{210}Pb separation

Conclusion

- It is necessary that the activity of ^{210}Pb of Pb^{2+} carrier is verified and blank-corrected for more high accuracy.
- It suggests that 1-Window method is more precise and reproducible with high detection efficiency (~180%) for the low-level environmental samples.
- 1-Window method has an advantage of using a small amount of sample in atmosphere, rainwater, and etc, due to relatively low minimum detectable activity (MDA).
- It will be expected to be used for the determination of other radionuclides with a short half-life daughter.